



H2O Online Platform

Complete configuration, deployment, API reference, and settings
guide

Backend · Mobile · Admin · Production

Generated 19 June 2026

H2O Online Platform

End-to-end water delivery platform: **mobile app** (customers, suppliers, delivery partners), **admin portal**, and **REST API** backed by MongoDB.

Component	Stack	Default port	Folder
Backend API	Node.js, Express 5, Mongoose	5000	backend/
Mobile app	Expo 54, React Native, Expo Router	8081 (Metro)	mobile/
Admin portal	Vite 5, React 18	5174 (dev) / 3000 (prod)	admin/
Documentation	Puppeteer (screenshots / PDF)	—	docs/

Table of contents

1. [Prerequisites](#)
2. [Quick start \(local\)](#)
3. [Project structure](#)
4. [Environment variables](#)
5. [Backend configuration](#)
6. [API reference](#)
7. [Admin portal](#)
8. [Mobile app](#)
9. [Production deployment](#)
10. [Domain & HTTPS](#)
11. [Documentation tooling](#)
12. [Troubleshooting](#)
13. [Security checklist](#)

Prerequisites

Tool	Version / notes
Node.js	18+ recommended
MongoDB	Local install or MongoDB Atlas
npm	Comes with Node
Expo CLI	For mobile dev (<code>npx expo</code>)
EAS CLI	For cloud APK builds (<code>npm install -g eas-cli</code>)
PM2	Production process manager (Ubuntu server)
Nginx	Optional reverse proxy + HTTPS

Quick start (local)

1. Backend

```
cd backend
npm install
cp .env.example .env          # edit MONGODB_URI, JWT_SECRET if needed
npm run check-mongo           # verify MongoDB is reachable
npm run seed                  # sample users, suppliers, products
npm run seed-plans            # subscription plans
npm run seed-delivery-partners
npm run dev                   # http://localhost:5000
```

Verify: open <http://localhost:5000/api/health> — expect `{ "ok": true, "db": "connected" }`.

2. Mobile app

```
cd mobile
npm install
npm run local                # auto-sets LAN IP in .env, starts Expo
```

Scan the QR code with **Expo Go** (phone and PC on the same Wi-Fi), or press `w` for web / `a` for Android emulator.

3. Admin portal

```
cd admin
npm install
cp .env.example .env          # VITE_API_URL=http://localhost:5000
npm run dev                   # http://localhost:5174
```

Master login: username `H2O admin` · password `admin@H2O`

Project structure

```
H2Oonline/
├─ backend/
│   ├── server.js           # Express entry point
│   ├── config/db.js        # MongoDB connection
│   ├── middleware/         # auth, adminAuth
│   ├── routes/             # REST route modules
│   ├── models/             # Mongoose schemas
│   ├── services/           # AI (Groq), payouts, reports
│   └─ scripts/             # seed, health checks
├─ mobile/
│   ├── app/                # Expo Router screens (51 routes)
│   ├── src/                # screens, API client, contexts
│   ├── config/
│   │   ├── api.js          # API URL resolution logic
│   │   └─ apiUrl.json      # Production API URL (single source)
│   ├── eas.json            # EAS build profiles + API URL for APK
│   └─ scripts/             # set-local-url, set-production-url, icons
├─ admin/
│   ├── src/pages/          # Dashboard, Users, Orders, Plans, etc.
│   └─ src/api/             # Admin API client
└─ docs/
    ├── capture-*.mjs        # Screenshot automation
    ├── generate-*.pdf.mjs    # PDF generation
    └─ DOMAIN-SETUP.md       # Nginx + Let's Encrypt guide
```

Environment variables

Backend — backend/.env

Copy from backend/.env.example :

Variable	Default	Description
PORT	5000	HTTP server port
MONGODB_URI	mongodb://127.0.0.1:27017/h2online	MongoDB connection string (local or Atlas mongodb+srv://...)
JWT_SECRET	*(change in production)*	Secret for signing JWT tokens (mobile + admin)
GROQ_API_KEY	—	Groq API key for Water AI features (insights, reports, ask)
GROQ_MODEL	llama-3.3-70b-versatile	Groq model name

The server binds to 0.0.0.0 so phones on the LAN can reach it. On startup it prints the LAN IP to use in mobile/.env .

Admin — admin/.env

Variable	Example	Description
VITE_API_URL	http://localhost:5000	Backend base URL — baked in at build time (Vite)

Production example: VITE_API_URL=http://13.62.57.255:5000 or https://api.yourdomain.com

Mobile — mobile/.env

Variable	Example	Description
EXPO_PUBLIC_API_URL	http://192.168.1.5:5000	Backend URL for Expo dev / Expo Go

How the mobile app picks the API URL (mobile/config/api.js):

- 1. EXPO_PUBLIC_API_URL from .env (highest priority)
- 2. Dev fallback: Android emulator → http://10.0.2.2:5000 , else → http://localhost:5000
- 3. Production APK fallback: mobile/config/apiUrl.json → production

Production APK URL is set in eas.json at build time — not from .env .

Backend configuration

NPM scripts

Command	Purpose
npm run dev	Start with nodemon (development)
npm start	Start production server
npm run seed	Seed users, suppliers, products
npm run seed-plans	Upsert subscription plans
npm run seed-delivery-partners	Sample delivery partners
npm run check-mongo	Test MongoDB connection
npm run check	Health + register smoke test

Additional scripts (run manually):

```
node scripts/seed-docs-demo.js      # Demo data for screenshot docs
node scripts/seed-pickup-hubs.js    # Pickup hub locations
node scripts/test-ai.js             # Test Groq AI endpoints
```

Database

- Database name: h20online
- Created automatically on first connection or seed
- Windows + Atlas: DNS fallback to 8.8.8.8 / 1.1.1.1 in config/db.js

Authentication

Audience	Header	Middleware
Mobile app users	Authorization: Bearer <token>	middleware/auth.js
Admin portal	Authorization: Bearer <token>	middleware/adminAuth.js

- JWT expiry: 7 days
- Default secret fallback: h20-secret (override with JWT_SECRET in production)

Seed credentials (development only)

Role	Email	Password
Supplier	aquapure@example.com	seedpass123
Delivery partner	rahul.dp@h2o.test	delivery123

API reference

Base URL: `http://HOST:5000`

Health checks:

Method	Path	Response
GET	/	"H2Online Backend Running"
GET	/api/health	{ ok, db, readyState }

Auth — /api/auth

Method	Path	Auth	Body / notes
POST	/register	—	Customer signup: { name, email, phone?, password, ... }
POST	/register-supplier	—	Supplier onboarding
POST	/register-delivery	—	Delivery partner signup
POST	/login	—	{ email, password } → { user, token }

Users — /api/users *(auth required)*

Method	Path
GET	/me
PUT	/me

Products — /api/products

Method	Path	Auth	Notes
GET	/	—	Query: search, minL, maxL, category, sort
GET	/:id	—	Single product
POST	/	auth	Create product
PUT	/:id	auth	Update product
DELETE	/:id	auth	Delete product

Orders — `/api/orders` ***(auth required)***

Method	Path
GET	<code>/</code>
GET	<code>/:id</code>
POST	<code>/</code>
PATCH	<code>/:id/cancel</code>

Reviews — `/api/reviews`

Method	Path	Auth
POST	<code>/</code>	auth
GET	<code>/product/:productId</code>	—
GET	<code>/me/order/:orderId</code>	auth

Wallet — `/api/wallet` ***(auth required)***

Method	Path
GET	<code>/</code>
POST	<code>/credit</code>
POST	<code>/debit</code>

Addresses — `/api/addresses` ***(auth required)***

Method	Path
GET	<code>/</code>
POST	<code>/</code>
PUT	<code>/:id</code>
DELETE	<code>/:id</code>

Plans — `/api/plans`

Method	Path	Auth
GET	<code>/</code>	—
GET	<code>/:slug/products</code>	—

Subscriptions — `/api/subscriptions` ***(auth required)***

Method	Path
GET	<code>/</code>
POST	<code>/</code>
PATCH	<code>/:id/cancel</code>

Bills — `/api/bills` ***(auth required)***

Method	Path
GET	<code>/</code>
POST	<code>/:id/pay</code>

Water intake — `/api/water-intake` ***(auth required)***

Method	Path
GET	<code>/</code>
POST	<code>/</code>
DELETE	<code>/:entryId</code>
GET	<code>/summary</code>

AI (Grog) — `/api/ai` ***(auth required, needs `GROQ_API_KEY`)***

Method	Path
GET	<code>/water-insight</code>
GET	<code>/intake-sense</code>
POST	<code>/water-report</code>
POST	<code>/ask</code>

Supplier — `/api/supplier` ***(auth required)***

Method	Path
GET	<code>/orders/incoming</code>
GET	<code>/orders/history</code>
GET	<code>/orders/accepted</code>
PATCH	<code>/orders/:id/accept</code>
PATCH	<code>/orders/:id/reject</code>
PATCH	<code>/orders/:id/assign-rider</code>
PATCH	<code>/orders/:id/cancel</code>
GET	<code>/products</code>
GET	<code>/financials</code>

Suppliers profile — `/api/suppliers` ***(auth required)***

Method	Path
GET	<code>/me</code>

Delivery partners — /api/delivery-partners

Method	Path	Auth
GET	/me	auth
PATCH	/me	auth
GET	/	auth
GET	/subscriptions	auth
GET	/orders/incoming	auth
GET	/orders/history	auth
GET	/orders/summary	auth
PATCH	/orders/:id/picked-up	auth
PATCH	/orders/:id/delivered	auth
GET	/financials	auth

Support

Prefix	Endpoints *(all auth)*
/api/customer-support	GET/POST /tickets , GET/POST /tickets/:id , POST /tickets/:id/reply
/api/supplier-support	GET /thread , POST /message
/api/delivery-support	GET /thread , POST /message

Admin — /api/admin

Auth:

Method	Path	Auth
POST	/auth/login	—
GET	/auth/me	adminAuth

Management *(all require adminAuth)*:

Area	Endpoints
Users	GET /users , GET /users/:id , PUT /users/:id , DELETE /users/:id
Admins	POST /admins
Orders	GET /orders , GET /orders/:id
Suppliers	GET/POST/PATCH/DELETE /suppliers , /suppliers/:id , /suppliers/:id/verify
Plans	GET /plans , PUT /plans/:id , CRUD /plan-products
Pickup hubs	GET/POST /pickup-hubs
Subscriptions	GET/PATCH/DELETE /subscriptions , assign delivery, financials
Wallet	GET /wallet-management , POST /wallet-management/:userId/adjust
Financials	GET /financials
Delivery partners	GET /delivery-partners , PATCH /delivery-partners/:id/verify
Support	Supplier, delivery, customer support threads and replies

Admin role permissions:

Role	Financials	Delete users	Remove suppliers	Create admin
Master	✓	✓	✓	✓
Admin	✓	✓	✓	✓
Sub-admin	✗	✗	✗	✗

Admin portal

Development

```
cd admin
npm install
cp .env.example .env
npm run dev
```

Open **http://localhost:5174**

Production build (Ubuntu / AWS)

```
cd admin
echo 'VITE_API_URL=http://YOUR_SERVER_IP:5000' > .env
npm install
npm run build
sudo npm install -g serve pm2
pm2 serve dist 3000 --spa --name "h20-admin"
pm2 save && pm2 startup
```

Open **http://YOUR_SERVER_IP:3000**

See also: `admin/RUN-ON-UBUNTU.md`

Admin UI routes

Path	Page
<code>/login</code>	Admin login
<code>/</code>	Dashboard
<code>/users</code>	User management
<code>/orders</code>	Orders
<code>/suppliers</code>	Suppliers
<code>/plans</code>	Plans & rates
<code>/subscriptions</code>	Subscriptions
<code>/wallet-management</code>	Wallet adjustments
<code>/financials</code>	Financials *(admin/master only)*
<code>/admin-users</code>	Create admin users *(admin/master only)*
<code>/delivery-partners</code>	Delivery partners
<code>/supplier-support</code>	Supplier support chat
<code>/delivery-support</code>	Delivery support chat
<code>/customer-support</code>	Customer tickets

Mobile app

App identity

Setting	Value
Display name	H2Oonline
Slug	<code>h2oonline</code>
Package / bundle ID	<code>com.h2oonline.app</code>
Scheme	<code>h2oonline</code>
EAS project ID	<code>617044f2-1ffb-421f-94e0-3d1cd4530c70</code>
EAS owner	<code>apps-sigma</code>
Cleartext HTTP	Enabled (<code>usesCleartextTraffic: true</code>)

NPM scripts

Command	Purpose
<code>npm run local</code>	Detect LAN IP → write <code>.env</code> → <code>expo start -c</code>
<code>npm run prod:local</code>	Point <code>.env</code> at production API → <code>expo start -c</code>
<code>npm run start</code>	Standard Expo start
<code>npm run web</code>	Expo web (port 8081)
<code>npm run android / ios</code>	Native dev build
<code>npm run icons</code>	Regenerate launcher icons from H2O logo
<code>npm run build:apk</code>	EAS production APK
<code>npm run build:apk:preview</code>	EAS preview APK

API URL modes

Mode	URL source	Command
Local dev (phone)	<code>mobile/.env</code> (auto LAN IP)	<code>npm run local</code>
Test production API in Expo	<code>mobile/.env</code> → AWS URL	<code>npm run prod:local</code>
Production APK	<code>eas.json</code> env at build time	<code>npm run build:apk</code>
Android emulator	Hardcoded <code>10.0.2.2:5000</code>	—

Change production API URL

Update **both** files, then rebuild the APK:

- `mobile/config/apiUrl.json` → `"production": "http://NEW_IP:5000"`
- `mobile/eas.json` → `EXPO_PUBLIC_API_URL` in each build profile

Current production value: `http://13.62.57.255:5000`

EAS build profiles (`eas.json`)

Profile	Distribution	Android	API URL
<code>development</code>	internal, dev client	—	AWS URL
<code>preview</code>	internal APK	<code>buildType: apk</code>	AWS URL
<code>production</code>	internal APK	<code>buildType: apk</code>	AWS URL

One-time EAS setup

```
npm install -g eas-cli
eas login
cd mobile
eas build:configure
```

Build production APK

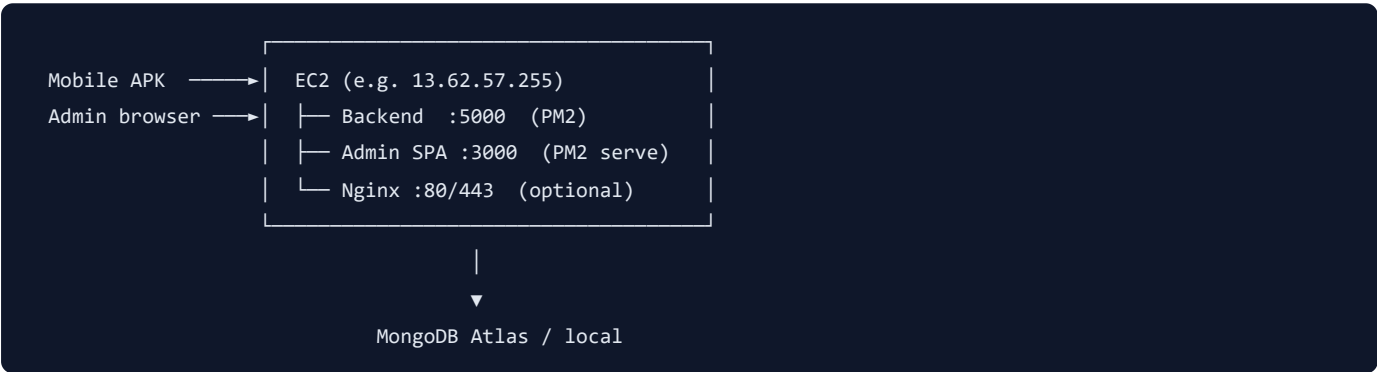
```
cd mobile
npm install
npm run icons           # once, or after logo change
npm run build:apk
```

See `mobile/BUILD_APK.md` for troubleshooting (EAS upload size, peer deps, cleartext).

Production deployment

Documented setup: **Ubuntu EC2** with PM2. No Docker configuration is included in this repo.

Architecture overview



Backend on Ubuntu

```
cd backend
npm install
cp .env.example .env
# Set MONGODB_URI (Atlas recommended for production)
# Set JWT_SECRET to a strong random value
# Set GROQ_API_KEY if using AI features
npm run seed && npm run seed-plans && npm run seed-delivery-partners
pm2 start server.js --name backend
pm2 save && pm2 startup
```

AWS security group (inbound rules)

Port	Protocol	Purpose
5000	TCP	Backend API
3000	TCP	Admin portal (if accessed directly)
80	TCP	HTTP (if using Nginx)
443	TCP	HTTPS (if using Nginx + Certbot)
22	TCP	SSH

Verify deployment

```
curl http://YOUR_SERVER_IP:5000/api/health
# Expected: {"ok":true,"db":"connected",...}
```

Domain & HTTPS

Full guide: [docs/DOMAIN-SETUP.md](#)

Summary:

1. **DNS A records** → point `api.yourdomain.com` and `admin.yourdomain.com` to your server IP
2. **Nginx** → proxy `api` → `127.0.0.1:5000`, serve admin from `admin/dist`
3. **Certbot** → `sudo certbot --nginx -d api.yourdomain.com -d admin.yourdomain.com`
4. **Rebuild admin** with `VITE_API_URL=https://api.yourdomain.com`
5. **Rebuild mobile APK** with `EXPO_PUBLIC_API_URL=https://api.yourdomain.com` in `eas.json`

Example Nginx API block:

```
server {
    listen 80;
    server_name api.yourdomain.com;
    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Documentation tooling

The `docs/` folder generates screenshot catalogs and PDFs for the platform.

```
cd docs
npm install
npm run docs          # Capture all app screens + PDF
npm run customer-journey # Customer-only journey PDF
```

Script	Output
<code>npm run capture</code>	Screenshots → <code>docs/screenshots/</code>
<code>npm run pdf</code>	<code>docs/H2O-Screens-Documentation.pdf</code>
<code>npm run customer-journey</code>	<code>docs/customer_journey.pdf</code>

Requires Expo web running on **http://localhost:8081** and backend on **:5000**.

Troubleshooting

Mobile cannot reach backend (timeout)

1. Backend running? Check `http://localhost:5000/api/health`
2. Phone and PC on the **same Wi-Fi**
3. Use PC **LAN IP** in `mobile/.env`, not `localhost`
4. Run `npm run local` in `mobile/` to auto-detect IP
5. **Windows Firewall** — allow inbound TCP 5000:

```
""powershell
```

```
New-NetFirewallRule -DisplayName "Node 5000" -Direction Inbound -LocalPort 5000 -Protocol TCP -Action Allow
```

```
""
```

1. **AWS:** security group must allow inbound TCP 5000

See `mobile/CONNECTION_HELP.md` and `mobile/LOCAL_SETUP.md`.

MongoDB connection failed

```
cd backend
npm run check-mongo
```

- Local: ensure MongoDB service is running
- Atlas: verify `MONGODB_URI`, IP whitelist, and network access

Admin shows blank or API errors after deploy

- `VITE_API_URL` must be the **public** URL the browser can reach (not `localhost` when opened from another machine)
- Rebuild after changing `.env`: `npm run build`
- Restart PM2: `pm2 restart h2o-admin`

EAS build fails (peer dependencies)

`mobile/.npmrc` sets `legacy-peer-deps=true`. Run `npm install` in `mobile/` before building.

AI features not working

Set `GROQ_API_KEY` in `backend/.env`. Test with `node scripts/test-ai.js`.

Security checklist

- ☐ Never commit `.env` files (`backend/.env`, `admin/.env`, `mobile/.env`)
- ☐ Set a strong `JWT_SECRET` in production
- ☐ Change master admin password or restrict admin access by IP
- ☐ Use MongoDB Atlas with IP whitelist in production
- ☐ Prefer HTTPS (Nginx + Certbot) over plain HTTP for production
- ☐ Rotate seed/demo passwords before going live
- ☐ Restrict AWS security group to required ports only
- ☐ Keep `GROQ_API_KEY` server-side only (never in mobile or admin client)

Related documentation

File	Contents
backend/README.md	Backend setup, seeding, firewall
admin/README.md	Admin roles and login
admin/RUN-ON-UBUNTU.md	PM2 deploy steps
mobile/BUILD_APK.md	EAS APK build guide
mobile/CONNECTION_HELP.md	Network troubleshooting
mobile/LOCAL_SETUP.md	Full local dev workflow
docs/DOMAIN-SETUP.md	Domain, Nginx, HTTPS

Key values quick reference

Setting	Value
Backend port	5000
Backend host	0.0.0.0
MongoDB database	h2online
Admin dev port	5174
Admin prod port	3000
Expo Metro port	8081
Production API	http://13.62.57.255:5000
Android emulator API	http://10.0.2.2:5000
Master admin login	H2O admin / admin@H2O
Mobile request timeout	25s (60s for AI endpoints)